

Phylogenetic Analysis Using NCBI and Galaxy

Group Number:

Group 3

Taxonomic Group:

Bovidae

Members:

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Course:

BIO 300: Cell and Molecular Biology

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Introduction

Phylogenetic analysis is a method used to determine the evolutionary relationships among organisms by comparing their genetic characteristics and identifying patterns of common ancestry. In this study, mitochondrial cytochrome c oxidase subunit I (COI or COX1) sequences are utilized because this gene is widely applied in DNA barcoding and evolutionary studies due to its high abundance in cells, maternal inheritance, and ability to provide sufficient genetic variation for distinguishing closely related species. The family Bovidae was chosen because it contains a diverse group of mammals, including cattle, buffaloes, goats, sheep, and antelopes, which display varying degrees of genetic relatedness and are suitable for examining evolutionary relationships through molecular analysis. The inclusion of *Bos taurus* as an outgroup provides a reference for comparing genetic differences and determining the evolutionary placement of the selected species. Therefore, this activity aims to construct and analyze a phylogenetic relationship among selected members of the family Bovidae using mitochondrial COI (COX1) gene sequences obtained from the NCBI database.

Biological Question

What are the evolutionary relationships among selected species of the family Bovidae based on mitochondrial cytochrome c oxidase subunit I (COI) gene sequences, using *Cervus elaphus* as the outgroup?

Based on mitochondrial COI (COX1) gene sequences, the selected species of the family Bovidae are expected to group according to their genetic similarities and evolutionary relationships. Species within the same genus, such as *Bos indicus* and *Bos taurus*, are expected to show the closest relationship due to their shared ancestry. *Bison bison* and *Syncerus caffer* may also cluster closely with cattle as members of the Bovidae group, while *Capra hircus* and *Ovis aries* are expected to form a separate group under Caprinae. *Cervus elaphus*, used as the outgroup, provides a reference for identifying evolutionary differences between Bovidae and a

more distantly related species. The analysis aims to reveal the phylogenetic relationships among these species based on COI sequence similarity.

I. Species Information

Scientific name	Common name	NCBI accession	Length (bp)	Focal taxon/ outgroup
<i>Bos indicus</i>	Zebu (Cattle)	PZ072327	665	Focal taxon
<i>Bison bison</i>	American Bison	JF443195	657	Focal taxon
<i>Antelope cervicapra</i>	Blackbuck	MT251370	657	Focal taxon
<i>Syncerus caffer</i>	Cape buffalo	JN082178	675	Focal taxon
<i>Capra hircus</i>	Goat	OK184394	658	Focal taxon
<i>Ovis aries</i>	Sheep	PX945640	592	Focal taxon
<i>Cervus elaphus</i>	Red deer	JF443224	657	Outgroup
<i>Bos taurus</i>	Cattle	MZ099227	660	Focal taxon

Table 1. Table containing list of species with their respective nucleotide accession number and bp length.

Methods

Sequence Retrieval

Mitochondrial cytochrome c oxidase subunit I (COI) sequences of selected Bovidae species were obtained from the NCBI Nucleotide database. Each species was searched individually using its scientific name together with the keywords "COI" or "COX1" and

"mitochondrion." Only mitochondrial COI sequences with similar lengths and covering the same gene region were selected to ensure accurate comparison. The sequences were downloaded in FASTA format. The red deer (*Cervus elaphus*) was selected as the outgroup for the phylogenetic analysis.

Data Preparation

The downloaded FASTA files were uploaded to the Galaxy platform. Each sequence was checked to ensure it was in FASTA format and renamed using short species labels for easier identification. All sequences were then combined into a single multi-FASTA file for alignment.

Multiple Sequence Alignment

The combined sequences were aligned using the MAFFT tool available in Galaxy. Default settings were used during the alignment. After the alignment was completed, the output was checked to confirm that the nucleotide positions were properly aligned and that there were no obvious errors or poor-quality sequences.

The screenshot shows the Galaxy web interface. The main panel displays the output of the MAFFT alignment tool. The title is "80: MAFFT on dataset 79" with a green "ok" status. Below the title are tabs for "Preview", "Visualize", "Details", and "Edit". The "Preview" tab is active, showing the aligned sequences for three species: *Bos taurus*, *Capra hircus*, and *Bison bison*. The sequences are aligned in a multi-FASTA format, with gaps represented by dashes (-). The right sidebar shows the history of datasets, including "80: MAFFT on dataset 79", "79: Bovidae_multi4.fasta", "78: FASTTREE on dataset 77: tree.nhx", and "77: MAFFT on dataset 76".

Figure 1. MAFFT alignment screenshot: Gaps were replaced with dash (-) to make every sequence the exact same length. This transforms the raw, unaligned sequences into uniform, co-linear alignment sequences.

Consensus Sequence Generation

A consensus sequence was generated from the aligned sequences using the consensus sequence tool in Galaxy with the default settings. The consensus sequence was examined to identify conserved and variable nucleotide sites among the selected species. This step helped summarize the common nucleotide pattern across the alignment. The consensus sequence was used only for sequence analysis and was not used to construct the phylogenetic tree.

Phylogenetic Tree Construction

The aligned sequences were used to construct a phylogenetic tree using the FastTree tool in Galaxy. The analysis was performed using nucleotide sequence data, and the resulting tree was generated in Newick format. The tree was then visualized using the Galaxy tree viewer, with the red deer (*Cervus elaphus*) serving as the outgroup to help interpret the evolutionary relationships among the Bovidae species.

Tree Interpretation

The phylogenetic tree was analyzed by identifying sister taxa, major clades, and the position of the outgroup. Branch lengths were also observed to compare the genetic differences among species. The resulting tree was then compared with published studies on the evolutionary relationships within the family Bovidae to determine whether the inferred relationships were consistent with previous research.

Galaxy Workflow and History

The screenshot displays the Galaxy web interface. The top navigation bar includes the Galaxy logo, a grid icon, a search icon, a home icon, a help icon, and a status bar indicating 'Using 0% of 268.4 GB' and the user 'shahantabalanza'. The left sidebar contains a vertical menu with icons for Activities, Visualization, Histories, History Multiview, Datasets, Reports, Libraries, Recent Exports, Notifications, and More. The main content area is titled 'Histories / Group_3_Bovidae current'. Below the title is a search bar labeled 'search datasets'. The history overview for 'Group_3_Bovidae' is shown, indicating 'Default Storage' and a size of '138 kB'. A list of history items is displayed, each with a checkbox, a zoom icon, and a settings icon. The items are: 83: Consensus.seqs on dataset 80: cons.fasta, 82: Consensus.seqs on dataset 80: cons.summary, 81: FASTTREE on dataset 80:tree.nhx, 80: MAFFT on dataset 79, 79: Bovidae_multi4.fasta, 78: FASTTREE on dataset 77:tree.nhx, 77: MAFFT on dataset 76, 76: Bovidae_mult3.fasta, and 75: FASTTREE on dataset 74:tree.nhx. Each item has a 'Switch to this history' button and a 'x' icon.

Figure 2. Galaxy analysis history overview for Group_3_Bovidae showing the complete workflow. The panel displays the user account, history title, uploaded multi-FASTA files, and the successive outputs generated from MAFFT alignment, FastTree phylogenetic tree construction, and consensus sequence generation.

Results

Multiple Sequence Alignment (MAFFT)

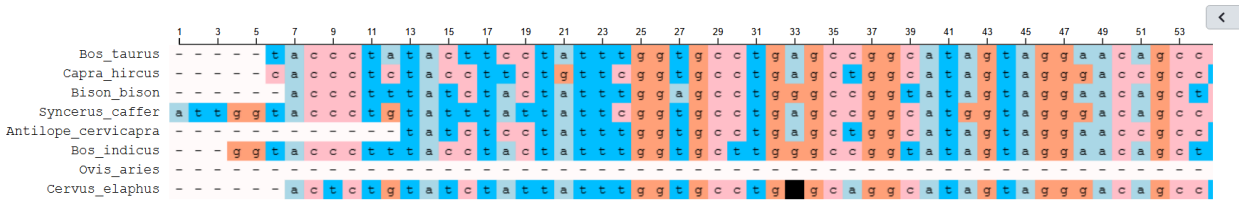


Figure 3. A screenshot of Representative MAFFT alignment.

Link:  **Multiple Sequence Alignment.pdf**

Consensus Sequence

Link: https://drive.google.com/file/d/17ceNwcZd3sz9J7Jg7CJibQJRrOX74WSq/view?usp=drive_sdk

The consensus sequence represents the most common nucleotide found at each position among the aligned COI sequences of the selected Bovidae species and the outgroup. It reflects the conserved regions shared by the species, indicating their common evolutionary ancestry. Positions that differ among the sequences are variable sites, which provide important information for distinguishing species and determining their evolutionary relationships. The consensus sequence summarizes the alignment but is not used to construct the phylogenetic tree; instead, the aligned sequences from all species are used for tree building.

Phylogenetic Tree

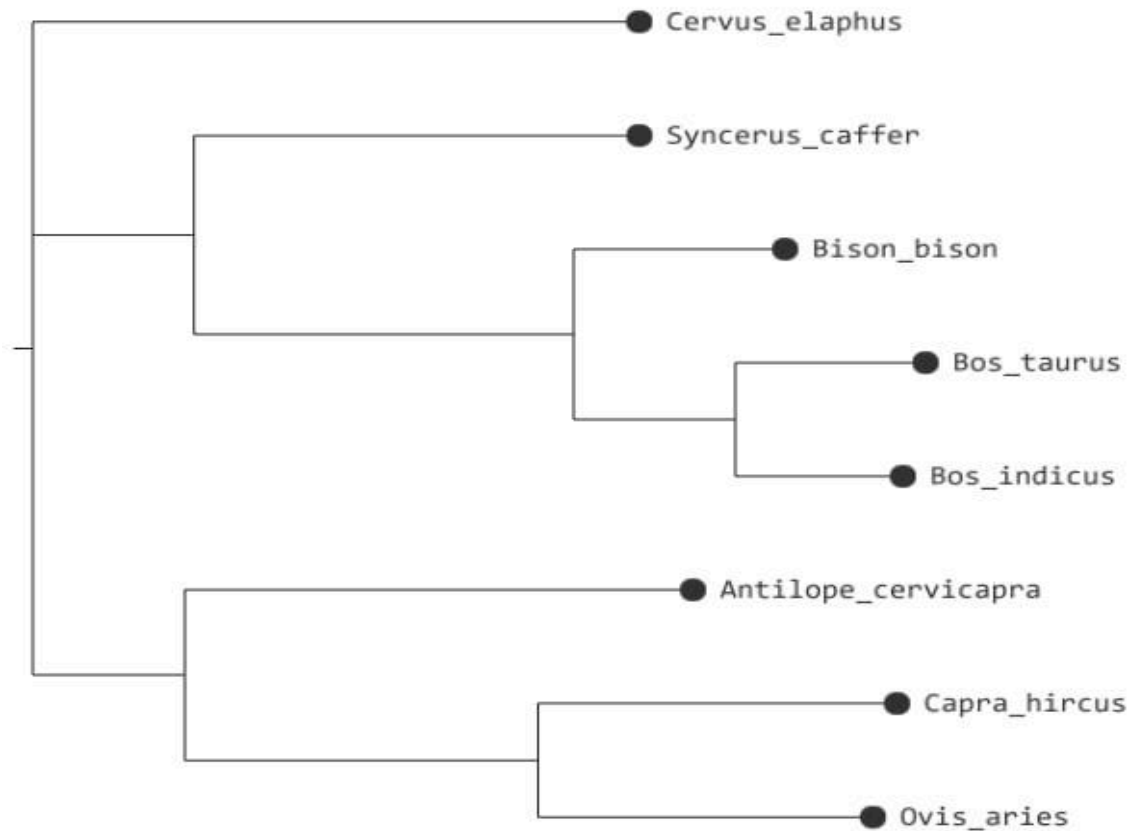


Figure 3. Phylogenetic tree of selected Bovidae species based on mitochondrial COI sequences.

The phylogenetic tree constructed from mitochondrial cytochrome c oxidase subunit I (COI) sequences shows the evolutionary relationships among selected members of the family Bovidae, with *Cervus elaphus* (family Cervidae) serving as the outgroup. The outgroup branches separately from all Bovidae, confirming that it is more distantly related and providing a reference for rooting the tree. Within the Bovidae, two major clades are evident. The first includes *Syncerus caffer*, *Bison bison*, *Bos taurus*, and *Bos indicus*, while the second consists of *Antilope cervicapra*, *Capra hircus*, and *Ovis aries*. The closest sister taxa are *Bos taurus* and *Bos indicus*, reflecting their recent common ancestry within the genus *Bos*. Likewise, *Capra hircus* and *Ovis aries* form another pair of closely related sister taxa within the subfamily Caprinae. *Bison bison* is closely related to the *Bos* species, whereas *Syncerus caffer* diverges earlier within the cattle

lineage. The branch lengths indicate the amount of genetic divergence among the COI sequences, with shorter branches representing closer genetic relationships. Overall, the tree is consistent with the accepted taxonomy of Bovidae. However, because it is based on a single mitochondrial gene, it should not be regarded as the definitive species phylogeny, as more comprehensive analyses generally incorporate multiple mitochondrial and nuclear genes to better resolve evolutionary relationships.

Comparison with Published Studies

Our COI phylogenetic tree is largely consistent with the molecular phylogeny reported by Hassanin et al. (2012). First, both trees recover *Bos taurus* and *Bos indicus* as sister taxa, reflecting their close evolutionary relationship within the genus *Bos* (Hassanin et al., 2012). Second, *Capra hircus* and *Ovis aries* are also grouped together as sister taxa in both phylogenies, supporting their shared ancestry within the subfamily Caprinae (Hassanin et al., 2012).

Our results also agree with the phylogenetic analysis of Bibi (2013). Similar to our tree, Bibi recovered the tribe Bovidae including *Bos*, *Bison*, and *Syncerus*, as a distinct evolutionary lineage within Bovidae. In both trees, *Bison bison* is more closely related to the *Bos* species than to goats or sheep, and *Syncerus caffer* belongs to the same broader cattle lineage (Bibi, 2013).

Discussion

One difference between our phylogeny and the published studies is the branching position of *Syncerus caffer* within the tribe Bovidae. Although our tree recovered *Syncerus caffer* as part of the Bovidae lineage, its exact branching position differs slightly from those reported by Bibi (2013) and Hassanin et al. (2012). These differences are likely due to the use of a single mitochondrial marker (COI) and a smaller number of taxa in our analysis, whereas the published studies included multiple mitochondrial and nuclear genes, larger taxon sampling, and different phylogenetic reconstruction methods.

References:

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